VSEPR-SIM THEORETICAL DIVISION / EVM PAPER SET

EVM-V

Galactic Rotation Residuals as Projection-Access Signals

Version 1.0

Abstract

This paper develops EVM-V, the first large-scale observational evidence-mapping document for the Inverse Kaluza-Klein Scale Symmetry framework. The preceding *Looking Up* paper proposed that galactic rotation anomalies may be interpreted not as direct evidence for unseen matter alone, nor as a purely empirical modification of Newtonian dynamics, but as a projection-residual effect associated with inaccessible scale-depth structure. In that formulation, the observed radial acceleration contains a baryonic Newtonian term plus an additional fifth-force-like contribution sourced by hidden-axis momentum and activated only at galactic scales.

The present paper does not claim discovery of a new force. Instead, it defines a reproducible data-analysis program for testing whether galactic residual accelerations can be mapped onto a projection-access model. The central empirical object is

$$a_\Pi(r) = a_{\rm obs}(r) - a_{\rm bar}(r),$$

where $a_{\rm obs} = v_{\rm obs}^2/r$ is the observed centripetal acceleration and $a_{\rm bar} = v_{\rm bar}^2/r$ is the acceleration predicted from baryonic matter alone. The paper treats the SPARC database as the primary benchmark, which contains 175 disk galaxies with *Spitzer* 3.6 μm photometry, high-quality H I/Hα rotation curves, and mass models suitable for baryonic acceleration reconstruction.

EVM-V compares three interpretations of the residual: conventional dark matter halo fitting, MOND-like phenomenological acceleration laws, and IKK projection-residual dynamics. The immediate goal is not to outperform ΛCDM or MOND, but to determine whether the projection-residual model yields stable population-level parameters, physically meaningful correlations, and non-arbitrary failure conditions. A model is considered scientifically useful only if its universal parameters remain stable across galaxies and its residual structure cannot be reduced to a disguised halo fit.

Keywords: galactic rotation curves; SPARC; radial acceleration relation; MOND; dark matter; fifth force; projection residual; Inverse Kaluza-Klein; scale symmetry; baryonic acceleration; hidden-axis momentum; effective force law.

§1 Introduction

The rotation-curve problem is one of the cleanest observational failures of naive baryonic Newtonian dynamics. In spiral and irregular disk galaxies, observed orbital velocities often remain approximately flat at large radii instead of declining as expected from visible mass alone. The standard interpretation is that galaxies are embedded in extended dark matter halos. This interpretation is mathematically productive and cosmologically powerful, but the underlying non-baryonic particle has not been directly detected. Modified gravity approaches, including MOND, instead treat the discrepancy as a modification of the acceleration law at low acceleration. The SPARC database and the radial acceleration relation have made this comparison unusually sharp because they connect observed acceleration and baryonic acceleration across thousands of resolved galactic radii.

The earlier *Looking Up* paper proposed a third interpretive route. Instead of treating the missing acceleration as either invisible mass or a primitive modification to gravity, it framed the anomaly as an unresolved projection residual. In the language of IKK, ordinary observation does not recover full physical identity:

$$\mathcal{O}_b^{(a)} = \mathcal{P}_{a\to b}[I_a], \qquad \mathcal{O}_b^{(a)} \neq I_a.$$

The observable object is a projection, not the primitive identity. The IKK notation registry explicitly preserves this distinction through the canonical rule

$$\mathfrak{I}_{\text{true}} \;\neq\; I_{\text{top}} \;\neq\; \mathcal{O}_b^{(a)}.$$

This distinction matters because a residual in galactic dynamics can be treated not as a free excuse term, but as the mismatch between gravitationally inferred structure and electromagnetically resolved structure.

The prior *Looking Up* model introduced hidden-axis momentum

$$p_w^{(i)} = m_i \,\frac{dw_i}{d\tau}$$

as the effective fifth-force charge, and proposed a logarithmic radial acceleration contribution which behaves as a $1/r$ acceleration at galactic radii, contributing an approximately constant term to $v^2(r)$.

EVM-V reframes this result as a data-analysis protocol, not a declaration of victory. Its central question is:

Can galactic acceleration residuals be mapped onto stable projection-access dynamics?

That is a useful question. It does not require pretending dark matter is dead, MOND is complete, or the fifth force has descended from the heavens carrying a grant proposal.

§2 Relation to *Looking Up*

The *Looking Up* paper developed a fifth-dimensional momentum framework for galactic rotation without invoking particle dark matter. Its abstract framed dark-sector signatures as "gravitationally visible but electromagnetically unresolved identity content," i.e. a projection gap rather than invisible matter.

The present paper differs in purpose.

Looking Up proposed a mechanism:

$$v^2(r) = v_N^2(r) + v_5^2(r).$$

EVM-V proposes an evidence pipeline:

data $\rightarrow$ $a_{\rm obs}$ $\rightarrow$ $a_{\rm bar}$ $\rightarrow$ a_{Π} $\rightarrow$ model comparison $\rightarrow$ parameter stability $\rightarrow$ failure conditions.

The paper therefore has a narrower and more scientific burden. It asks whether the fifth-force-like term from *Looking Up* can survive direct comparison against SPARC rotation curves, the radial acceleration relation, MOND-like scaling, and conventional dark matter halo fits.

The earlier paper already identified the correct primary benchmark: SPARC rotation-curve fitting, radial acceleration relation analysis, compact-system screening tests, population tests, and morphological correlations. EVM-V expands those into a full research design.

§3 Notation and Framework Alignment

The notation used here follows the IKK registry where possible.

3.1 Identity and projection

Let I_a denote primitive or source identity at scale S_a. Observation at scale S_b produces

$$\mathcal{O}_b^{(a)} = \mathcal{P}_{a\to b}[I_a],$$

where $\mathcal{P}_{a\to b}$ is the projection operator from source scale S_a into observable scale S_b. The registry gives the canonical operational form as

$$\mathcal{P}_{a\to b} = \mathcal{M}_b \circ \mathcal{R}_b \circ \mathcal{C}_{ab},$$

or, in simple form,

$$\mathcal{P}_{a\to b} = \mathcal{R}_b \circ \mathcal{C}_{ab},$$

where $\mathcal{C}_{ab}$ is the cross-scale coupling rule and $\mathcal{R}_b$ is the resolution operator of observing scale S_b.

3.2 Scale coordinate

The legacy coordinate w appears in *Looking Up* because that paper was written before the current notation registry stabilised. In this paper, the preferred coordinate is

$$u = \ln\left(\frac{\ell}{\ell_0}\right),$$

where ℓ is the relevant physical scale and ℓ_0 is a reference scale. The registry marks u as the accepted logarithmic scale coordinate and marks w as deprecated in new writing.

To preserve continuity with *Looking Up*, we write p_u as the modernised analogue of hidden-axis momentum, with the understanding that it descends from the earlier p_w construction:

$$p_w \longrightarrow p_u.$$

The replacement is conceptual as well as symbolic. The fifth direction is no longer treated as an ordinary hidden spatial coordinate. It is a scale-access coordinate governing how identity content becomes dynamically visible or invisible under projection.

3.3 Residual acceleration

The observable acceleration residual is defined as

$$a_{\Pi}(r) = a_{\text{obs}}(r) - a_{\text{bar}}(r).$$

The symbol Π is used here only as a typed residual label, not as a generic projection operator. The notation registry explicitly warns against bare Π because of projection and chemistry collisions. Tiny symbol hygiene, gigantic reduction in future suffering.

§4 Physical Hypothesis

The working hypothesis of EVM-V is:

$$a_{\text{obs}}(r) = a_{\text{bar}}(r) + a_{\Pi}(r, u),$$

where $a_{\Pi}(r, u)$ is an effective projection-residual acceleration.

This is not initially claimed to be a fundamental force in the Standard Model sense. It is a fifth-force-like term in the limited phenomenological sense that it contributes an additional acceleration:

$$F_{\text{eff}} = m\, a_{\text{obs}} = m\, a_{\text{bar}} + m\, a_{\Pi}.$$

The interpretation is:

> *The residual is observable; the ontology of the residual is inferred.*

This is the central scientific posture of the paper. The residual acceleration can be measured from rotation-curve and baryonic-mass data. The claim that it reflects projection-access structure is a model interpretation, and that interpretation must earn credibility through parameter stability, cross-galaxy consistency, and comparison with existing alternatives.

§5 Empirical Baseline: Rotation Curves and the RAR

5.1 Observed acceleration

For each galaxy, the observed centripetal acceleration at radius r_j is

$$a_{\text{obs}}(r_j) = \frac{v_{\text{obs}}^2(r_j)}{r_j}.$$

The observational inputs are radius, observed velocity, and velocity uncertainty:

$$\{r_j,\ v_{\text{obs}}(r_j),\ \sigma_v(r_j)\}_{j=1}^{N}.$$

5.2 Baryonic acceleration

The baryonic velocity contribution is decomposed as

$$v_{\text{bar}}^2(r) = v_{\text{gas}}^2(r) + \Upsilon_{\text{disk}}\, v_{\text{disk}}^2(r) + \Upsilon_{\text{bulge}}\, v_{\text{bulge}}^2(r),$$

where Υ_{disk} and Υ_{bulge} are stellar mass-to-light ratios. The baryonic acceleration is

$$a_{\text{bar}}(r) = \frac{v_{\text{bar}}^2(r)}{r}.$$

This is why SPARC is the natural first benchmark: it provides rotation curves and baryonic mass models for 175 disk galaxies across wide ranges of stellar mass, surface brightness, and gas fraction.

5.3 Radial acceleration relation

The empirical radial acceleration relation connects a_{obs} and a_{bar} across resolved radii in disk galaxies. McGaugh, Lelli, and Schombert reported a tight relation using SPARC data, making it one of the most direct targets for any projection-residual theory.

The EVM-V model must therefore be tested against both individual rotation curves and the population-level relation

$$a_{\text{obs}} = f(a_{\text{bar}}).$$

A model that fits individual galaxies but destroys the radial acceleration relation is not a success. It is just a mathematical ferret loose in the data.

§6 MOND as Phenomenological Baseline

MOND provides the most relevant phenomenological comparison because it already treats the anomaly as an acceleration-scale effect. In the deep-MOND regime,

$$a_{\text{MOND}} \approx \sqrt{a_0\, a_{\text{bar}}},$$

where a_0 is the characteristic acceleration scale.

The corresponding MOND residual is

$$a_{\Pi,\text{MOND}}(r) = a_{\text{MOND}}(r) - a_{\text{bar}}(r).$$

EVM-V uses this not as an assumption, but as a benchmark. The IKK projection-residual model should reproduce MOND-like behaviour in the regime where MOND is empirically successful:

$$a_{\Pi}(r,u) \approx a_{\Pi,\text{MOND}}(r) \quad \text{when } a_{\text{bar}} \ll a_0.$$

The interpretation differs:

MOND describes the low-acceleration law.

IKK attempts to explain *why* a scale-access law appears.

This is the correct relationship. EVM-V is not "MOND but with extra mist." It is a proposed ontological mapping beneath MOND-like phenomenology. Whether that mapping is worth anything depends on the analysis.

Li et al. fitted the radial acceleration relation to individual SPARC galaxies and found that allowing galaxy-to-galaxy variation in the acceleration scale did not improve the fits, which strengthens the burden on any IKK-like model: any additional scale-access parameter must either reduce to a nearly universal value or be independently tied to observable structure.

§7 Projection-Residual Model

7.1 Minimal residual form

The minimal EVM-V acceleration model is

$$a_{\text{obs}}(r) = a_{\text{bar}}(r) + a_{\Pi}(r,u).$$

A direct descendant of *Looking Up* is

$$a_{\Pi}(r,u) = \Lambda_{\Pi}\, \ln\!\left(1 + \frac{|p_u(r)|}{p_0}\right) \frac{G_{\Pi}(r)}{r + r_c},$$

where

$$G_\Pi(r) = \frac{1}{1 + e^{-k(r-r_0)}}.$$

This gives the full form:

$$a_\Pi(r,u) = \Lambda_\Pi \ln\left(1 + \frac{|p_u(r)|}{p_0}\right) \cdot \frac{1}{1 + e^{-k(r-r_0)}} \cdot \frac{1}{r + r_c}.$$

The corresponding velocity contribution is

$$v_\Pi^2(r) = r\, a_\Pi(r,u),$$

so that

$$v_{\mathrm{model}}^2(r) = v_{\mathrm{bar}}^2(r) + v_\Pi^2(r).$$

This is the modern-notation version of the *Looking Up* rotation-curve equation, where the fifth-force velocity term was written with Λ_5, p_w, p_0, r_0, k, and r_c.

7.2 Outer-disk limit

At large galactic radii, $r \gg r_0$ and $r \gg r_c$, so $G_\Pi(r) \to 1$ and $r/(r + r_c) \to 1$. Then

$$v_\Pi^2(r) \approx \Lambda_\Pi \ln\left(1 + \frac{|p_u(r)|}{p_0}\right).$$

If $p_u(r)$ varies slowly across the outer disk, v_Π^2 is approximately constant, producing flat rotation curves. This is exactly the key result of the original *Looking Up* derivation.

7.3 Projection-access interpretation

The term $p_u(r)$ is not directly observed. It is a proxy for scale-access momentum or hidden projection activity. Formally,

$$p_u(r) \sim \frac{\partial I_a}{\partial u},$$

meaning it measures how strongly the identity content changes along the scale-access coordinate. This should not be oversold. It is a modelling bridge, not a measured microscopic quantity.

Define a projection-access efficiency $\eta_\Pi(r,u) \in [0,1]$, and write the residual more generally as

$$a_\Pi(r,u) = \eta_\Pi(r,u)\, a_s(r),$$

where $a_s(r)$ is the scale-residual acceleration kernel. The logarithmic fifth-momentum model is then one candidate specification:

$$\eta_\Pi(r,u)\, a_s(r) = \Lambda_\Pi \ln\left(1 + \frac{|p_u(r)|}{p_0}\right) G_\Pi(r)\, \frac{1}{r + r_c}.$$

The notation registry currently treats η_{ab} as a provisional projection efficiency or coupling strength, which fits this use but requires explicit definition in this paper.

§8 Hidden-Momentum Proxy Models

The single largest modelling risk is $p_u(r)$. If it is freely chosen per galaxy, the model degenerates into a disguised halo fit. The prior *Looking Up* paper already identified this as the hidden-momentum proxy problem.

EVM-V therefore defines four proxy tiers.

Tier I: Circular-kinematic proxy

$$|p_u(r)| = \alpha_u\, M_\star\, v_{\rm circ}(r).$$

Simplest model. Ties scale-access momentum to local circular motion. Risks circularity because $v_{\rm circ}$ is close to the observable being fitted. Use only as a baseline sanity check.

Tier II: Surface-density proxy

$$|p_u(r)| = \alpha_u\, \Sigma_{\rm bar}(r)\, v_{\rm bar}(r).$$

Here $\Sigma_{\rm bar}(r)$ is the baryonic surface density. Ties the residual to local baryonic structure. This is the preferred first serious proxy.

Tier III: Momentum-flux proxy

$$|p_u(r)| = \alpha_u\, \rho_{\rm bar}(r)\, v_{\rm bar}(r).$$

Treats hidden-axis sourcing as related to local baryonic momentum density. More physically motivated but requires assumptions about vertical disk structure and density reconstruction.

Tier IV: Morphology-conditioned proxy

$$|p_u(r)| = \alpha_u\, f(\Sigma_{\rm bar},\, M_\star,\, f_{\rm gas},\, R_d,\, \text{morph.})\, v_{\rm bar}(r).$$

Should be used only after Tiers I–III are evaluated. Otherwise the model will do what models love doing when unsupervised: eat degrees of freedom and call it insight.

§9 Data Sources

9.1 Primary dataset: SPARC

The primary dataset is SPARC, which contains 175 nearby late-type galaxies with *Spitzer* 3.6 μm photometry and high-quality H I/Hα rotation curves. Required SPARC fields per data point:

$$r_j,\ v_{\rm obs}(r_j),\ \sigma_v(r_j),\ v_{\rm gas}(r_j),\ v_{\rm disk}(r_j),\ v_{\rm bulge}(r_j),\ L_{3.6},\ D,\ i.$$

Here D is distance and i is inclination.

9.2 Derived quantities

For each data point:

$$a_{\rm obs}(r_j) = \frac{v^2_{\rm obs}(r_j)}{r_j}, \tag{1}$$

$$v^2_{\rm bar}(r_j) = v^2_{\rm gas}(r_j) + \Upsilon_{\rm disk}\, v^2_{\rm disk}(r_j) + \Upsilon_{\rm bulge}\, v^2_{\rm bulge}(r_j), \tag{2}$$

$$a_{\rm bar}(r_j) = \frac{v^2_{\rm bar}(r_j)}{r_j}, \tag{3}$$

$$a^{\rm data}_{\Pi}(r_j) = a_{\rm obs}(r_j) - a_{\rm bar}(r_j). \tag{4}$$

9.3 Velocity uncertainty propagation

Velocity uncertainty propagates to acceleration as

$$\sigma_{a_{\mathrm{obs}}} = \frac{2v_{\mathrm{obs}}\,\sigma_v}{r}.$$

Uncertainty from stellar mass-to-light ratios Υ_{disk}, $\Upsilon_{\mathrm{bulge}}$ must also be included as systematic floors on a_{bar}.

9.4 Future dataset expansion

BIG-SPARC has been proposed as a much larger successor database, targeting approximately 4000 galaxies using H I data cubes from public telescope archives and near-infrared photometry from WISE. EVM-V begins with SPARC because it is established and manageable. BIG-SPARC belongs in EVM-VI or a later population-scale validation paper, assuming the model survives initial contact with reality. Harsh environment, reality.

§10 Analysis Pipeline

10.1 Preprocessing

For each galaxy:

1. Load SPARC rotation-curve and mass-model data.
2. Remove flagged or low-quality points according to SPARC quality metadata.
3. Convert radii to SI units or use consistent astrophysical units (kpc, $\mathrm{km\,s^{-1}}$, $\mathrm{m\,s^{-2}}$).
4. Compute v_{bar}, a_{obs}, a_{bar}, and a_{Π}^{data}.
5. Propagate velocity uncertainties to $\sigma_{a_{\mathrm{obs}}}$.
6. Include uncertainty from mass-to-light ratios Υ_{disk}, $\Upsilon_{\mathrm{bulge}}$ as systematic floors on a_{bar}.

10.2 Baseline residual extraction

Compute

$$\Delta v^2(r) = v_{\mathrm{obs}}^2(r) - v_{\mathrm{bar}}^2(r)$$

and

$$a_{\Pi}^{\mathrm{data}}(r) = \frac{\Delta v^2(r)}{r}.$$

This gives the empirical residual without assuming IKK, MOND, or halo structure.

10.3 Model fitting

Fit

$$v_{\mathrm{model}}^2(r;\Theta) = v_{\mathrm{bar}}^2(r) + \Lambda_\Pi\, r\, \ln\!\left(1 + \frac{|p_u(r)|}{p_0}\right) G_\Pi(r)\, \frac{1}{r + r_c}.$$

Parameter vector:

$$\Theta = \{\Lambda_\Pi,\ p_0,\ r_c,\ r_0,\ k,\ \alpha_u\}.$$

Depending on proxy tier, α_u may be universal, galaxy-class-dependent, or absorbed into p_0. Minimise

$$\chi^2(\Theta) = \sum_{j=1}^{N} \left[\frac{v_{\rm obs}(r_j) - v_{\rm model}(r_j;\,\Theta)}{\sigma_v(r_j)} \right]^2 .$$

Also compute acceleration-space loss:

$$\chi^2_a(\Theta) = \sum_{j=1}^{N} \left[\frac{a_{\rm obs}(r_j) - a_{\rm model}(r_j;\,\Theta)}{\sigma_{a,j}} \right]^2 .$$

Both should be reported because velocity-space and acceleration-space fitting can weight data differently.

10.4 Comparison models

For each galaxy, compare against:

1. **Newtonian baryons only**: $v^2_{\rm model} = v^2_{\rm bar}$.
2. **MOND/RAR**: $a_{\rm MOND} = \nu(a_{\rm bar}/a_0)\, a_{\rm bar}$, with a chosen interpolation function ν.
3. **Dark matter halo**: At minimum NFW, $\rho_{\rm NFW}(r) = \rho_s\big[(r/r_s)(1 + r/r_s)^2\big]^{-1}$; optionally Burkert or Einasto profiles.
4. **IKK projection-residual**: $v^2_{\rm model} = v^2_{\rm bar} + v^2_{\Pi}$.

10.5 Model-selection metrics

Report χ^2_ν, AIC, BIC, ΔAIC, ΔBIC, where

$$\mathrm{AIC} = 2k - 2\ln\mathcal{L}, \qquad \mathrm{BIC} = k\ln N - 2\ln\mathcal{L}.$$

This matters because the projection model must not win merely by having more knobs. Humanity already invented epicycles once. We do not need a premium subscription version.

§11 Population-Level Tests

A galaxy-by-galaxy fit is insufficient. The decisive test is whether the model has stable population-level structure.

11.1 Universal parameter stability

The parameters Λ_Π, p_0, r_c should be approximately universal or should vary according to a physically justified predictor. Define the population stability statistic:

$$S_\theta = \frac{\mathrm{std}[\log_{10}\theta_g]}{\sigma_{\theta,\rm allowed}},$$

where θ_g is the fitted value for galaxy g. A parameter passes if $S_\theta \le 1$. A practical criterion is

$$\max_g(\theta_g)\,/\,\min_g(\theta_g) < 3 \quad \text{(strong stability)}$$

and < 10 for weak admissibility. Beyond that, the model starts wearing a fake halo moustache.

11.2 Correlation with galaxy properties

If parameters vary, variation must correlate with observables: $M_\star$, $M_{\rm gas}$, $f_{\rm gas}$, R_d, Σ_0, surface brightness, morphology. For each parameter θ, test:

$$\log\theta = \beta_0 + \beta_1 \log M_\star + \beta_2 \log f_{\rm gas} + \beta_3 \log R_d + \epsilon.$$

No correlation plus high variation means failure.

11.3 RAR preservation

The projection-residual model must preserve the observed tightness of the RAR. Define residuals:

$$\delta_{\rm RAR} = \log a_{\rm obs} - \log a_{\rm model}.$$

Report $\sigma_{\rm RAR} = \mathrm{std}(\delta_{\rm RAR})$. A serious IKK model should approach the scatter achieved by empirical RAR/MOND-style fits. Li et al. found small residual scatter when fitting the RAR to SPARC galaxies, so EVM-V must treat that as a high bar, not a decorative citation.

§12 Screening and Compact-System Consistency

The original *Looking Up* model uses a radial sigmoid gate

$$G_\Pi(r) = \frac{1}{1 + e^{-k(r-r_0)}}$$

to suppress the fifth-force-like contribution below galactic scales. EVM-V must turn this into a testable consistency requirement.

12.1 Solar-system constraint

For solar-system radii, $r \ll r_0$, so $G_\Pi(r) \approx 0$. The model must satisfy

$$|a_\Pi(r_{\rm solar})| \ll a_{\rm bound},$$

where $a_{\rm bound}$ is the tightest available experimental bound on anomalous accelerations at solar-system scales.

12.2 Wide-binary and dwarf-galaxy transition zone

The danger zone is not molecular dynamics. The danger zone is intermediate gravitational systems: $r \sim 10^{-3}$ to 10^1 kpc. The model must specify whether the transition is determined by absolute radius, system mass, acceleration scale, environmental density, or scale coordinate u.

A pure radius gate may be too crude. A more defensible gate is $G_\Pi = G_\Pi(u, a_{\rm bar}, \Sigma_{\rm bar})$ rather than $G_\Pi = G_\Pi(r)$ alone. A radius-only gate is mathematically convenient but physically suspicious. Nature rarely respects our desire for easy logistic functions, rude little universe that it is.

§13 Cluster-Scale Consistency

Galaxy clusters are the hardest test for modified-gravity and fifth-force-like frameworks. The *Looking Up* paper explicitly notes that the Bullet Cluster creates a major constraint because the lensing mass follows the collisionless galaxies rather than the X-ray gas, even though the gas contains much of the baryonic mass.

EVM-V therefore requires a separate cluster analysis. The model must explain why a_Π^{cluster}, or the equivalent lensing residual, follows collisionless stellar components rather than collisional gas.

Possible mechanisms include:

- p_u is sourced primarily by collisionless stellar or compact-object components.
- Gas thermalisation suppresses coherent scale-access momentum.
- Projection residuals couple to identity persistence rather than raw baryonic mass.
- The IKK residual explains galaxy rotation curves but not cluster lensing, requiring ordinary dark matter or another sector at cluster scales.

The fourth option must remain allowed. Better an honest hybrid model than a grand unified failure balloon.

§14 Failure Conditions

EVM-V must define failure conditions before fitting the data. Otherwise the analysis becomes astrology with better typography.

Definition 14.1 (Weakened)**.** *The model is* weakened *if* Λ_Π, p_0, r_c *vary by more than an order of magnitude across galaxies without correlation to physical observables.*

Definition 14.2 (Strongly weakened)**.** *The model is* strongly weakened *if* $\mathrm{BIC}_{\mathrm{IKK}} > \mathrm{BIC}_{\mathrm{MOND}}$ *and* $\mathrm{BIC}_{\mathrm{IKK}} > \mathrm{BIC}_{\mathrm{NFW}}$ *for most galaxies despite having comparable or greater parameter freedom.*

Definition 14.3 (Rotation-model falsification)**.** *The model is* falsified as a useful first-order rotation model *if* $v_{\text{model}}(r)$ *cannot match SPARC rotation curves at the level achieved by standard halo models even with more parameters.*

Definition 14.4 (Descriptive curve fitting)**.** *The model is* reclassified as descriptive curve fitting *if* $p_u(r)$ *must be independently hand-shaped for every galaxy with no population-level structure.*

Definition 14.5 (Theoretically weakened)**.** *The model is* theoretically weakened *if* $G_\Pi(r)$ *must be manually adjusted to avoid solar-system and compact-system constraints rather than emerging from the scale interval structure of IKK IV.*

Definition 14.6 (Cluster-incomplete)**.** *The model is* cluster-incomplete *if it cannot explain lensing offsets in merging clusters without invoking a completely separate dark matter sector with no principled boundary from the galactic sector.*

These are not admissions of defeat. They are guardrails. Without them, the theory becomes immortal in the worst possible way: impossible to kill because it never actually stands anywhere.

§15 Proposed Figures and Tables

Figure 1 — EVM-V inference chain.

$$v_{\text{obs}}(r) \;\rightarrow\; a_{\text{obs}}(r) \;\rightarrow\; a_{\text{bar}}(r) \;\rightarrow\; a_\Pi(r) \;\rightarrow\; \text{IKK / MOND / halo comparison.}$$

Figure 2 — Projection-residual interpretation. Show $\mathfrak{I}_{\text{true}} \to \mathcal{P}_{a\to b} \to \mathcal{O}_b^{(a)}$ with a residual branch $\mathfrak{R}_{a|b} = \mathfrak{I}_{\text{true}} - \widehat{I}_{a|b}$.

Figure 3 — Rotation curve decomposition. Plot v_{obs}, v_{bar}, v_{Π}, v_{model} for representative SPARC galaxies at a range of stellar masses and surface brightnesses.

Figure 4 — RAR comparison. Plot a_{obs} vs. a_{bar} with curves for the empirical RAR, MOND, and the IKK residual model overlaid.

Figure 5 — Parameter stability. Plot fitted Λ_{Π}, p_0, and r_c across all galaxies with bootstrap confidence intervals, ordered by stellar mass.

Table 1: Model comparison summary

Model	**Free params**	**Data inputs**	**Main strength**	**Main failure mode**
Baryons only	0–2	SPARC mass model	Minimal	Fails outer curves
MOND/RAR	1–2	a_{bar}	Tight empirical law	Cluster/cosmology stress
NFW halo	2+	Halo profile	Cosmology-compatible	Halo–baryon tuning
IKK residual	4–6	Baryons + p_u proxy	Ontological residual map	Proxy degeneracy

Table 2: Hidden-momentum proxy hierarchy

Tier	**Proxy**	**Risk**	**Recommended use**
I	$M_\star v_{\mathrm{circ}}$	Circularity	Baseline only
II	$\Sigma_{\mathrm{bar}} v_{\mathrm{bar}}$	Surface-density assumptions	First serious model
III	$\rho_{\mathrm{bar}} v_{\mathrm{bar}}$	Vertical structure assumptions	Advanced
IV	Morphology-conditioned	Overfitting	Only after Tier II/III

§16 Minimal Reproducible Computational Workflow

A first implementation should produce the following files per galaxy:

```
galaxy_id/
  input/
    sparc_rotation_curve.csv
    sparc_mass_model.csv
  derived/
    acceleration_table.csv
    residual_table.csv
  fits/
    baryon_only_fit.json
```

```
    mond_fit.json
    nfw_fit.json
    ikk_projection_fit.json
  figures/
    rotation_curve_decomposition.png
    acceleration_relation.png
    residual_profile.png
  report/
    galaxy_summary.md
```

Minimum table schema (one row per resolved radius):

```
r_kpc, v_obs_kms, sigma_v_kms,
v_gas_kms, v_disk_kms, v_bulge_kms, v_bar_kms,
a_obs, a_bar, a_residual,
p_u_proxy, v_model_ikk, v_model_mond, v_model_nfw
```

Population-level outputs:

```
population_parameter_stability.csv
model_comparison_AIC_BIC.csv
rar_residuals.csv
morphology_correlation_summary.csv
failure_flags.csv
```

The most important file is `failure_flags.csv`, because apparently someone has to protect the theory from its own optimism.

§17 Discussion

EVM-V does not prove the IKK interpretation. Its contribution is methodological. It converts the *Looking Up* framework from a theoretical proposal into a structured observational programme.

17.1 Outcome classification

Strongest outcome. The model fits SPARC rotation curves competitively, Λ_Π, p_0, and r_c remain approximately stable (variation within a factor of 3 across the sample), the RAR scatter is preserved or reduced relative to the MOND baseline, and the cluster transition can be located at a principled scale boundary. In this outcome, the projection-residual interpretation earns the right to be taken seriously as a candidate framework alongside MOND and ΛCDM. It does not win by default, but it passes the minimum bar for scientific credibility.

Intermediate outcome. The model fits most SPARC galaxies acceptably, but Λ_Π, p_0, r_c show structured variation correlated with galaxy mass, gas fraction, or surface brightness. In this case the model is a useful phenomenological classifier: it maps different galaxy populations onto different projection regimes and predicts where the residual is large, small, or anomalous. This outcome is scientifically interesting but requires an explanation of why projection efficiency varies with observable morphological properties. It establishes population-level phenomenology that may or may not reflect a deeper mechanism.

Weak outcome. Parameters vary by more than an order of magnitude without physical correlation. Fitting is acceptable for individual galaxies but requires galaxy-by-galaxy p_u tuning equivalent to choosing a halo profile. In this case the IKK residual model has not added anything over standard halo fitting. It has only added vocabulary.

Falsification outcome. The model cannot reproduce SPARC rotation curves at the level of NFW halo fits even with more parameters, or BIC is substantially worse despite equal parameter freedom. This outcome would demote EVM-V from a projection-access programme to a falsified application. It would not falsify the abstract IKK projection framework — which is structural, not tied to any one dynamical application — but it would sever the *Looking Up* bridge from the formal theory to astrophysical data.

17.2 Relationship to MOND

The projection-residual framework makes a different ontological claim than MOND, but it must reproduce MOND-like phenomenology in the regimes where MOND succeeds. The natural consistency requirement is that the outer-disk flat-rotation limit of the IKK model recovers the deep-MOND scaling $a_{\mathrm{MOND}} \approx \sqrt{a_0 a_{\mathrm{bar}}}$. Whether this requires fine-tuning of Λ_Π and p_0 to match a_0, or whether it emerges naturally from the projection-access structure, is a key diagnostic. If the former, the model is a reparametrisation of MOND rather than an explanation of it.

17.3 Relationship to dark matter

Halo fits introduce spatially distributed mass profiles. Projection residuals introduce a scale-access coupling that does not require extra mass. The two frameworks are phenomenologically distinguishable only where they predict different radial acceleration profiles, RAR scatter patterns, or cluster-scale behaviour. EVM-V is specifically designed to identify those distinguishing observables. If no such observable can be found at the SPARC level, the two frameworks are empirically degenerate at galactic rotation-curve scales, and the competition must move to cluster lensing, galaxy formation statistics, or direct detection programmes.

17.4 Fifth-force interpretation and theoretical coherence

The term $a_\Pi(r, u)$ is described throughout as a "fifth-force-like" contribution. This language is deliberately cautious. The IKK framework does not introduce a new force in the gauge-theory sense. It introduces a projection-access coupling that produces an effective force at scales where the coupling is non-negligible. The distinction matters: a gauge force requires a mediator particle and a conserved charge detectable in particle physics. A projection coupling requires neither, but it also provides fewer experimental handles.

The theoretical coherence requirement is that $a_\Pi(r, u)$ must connect smoothly to the formal projection operator machinery of IKK I–IV. Specifically, $p_u(r)$ must be expressible as a functional of the projection access efficiency η_{ab}, and the $G_\Pi(r)$ gate must be derivable from the scale interval structure of IKK IV rather than chosen purely for curve-fitting convenience. Achieving that connection is a later-stage theoretical task; EVM-V opens the empirical space that makes such a connection worth pursuing.

17.5 The proxy problem is the central risk

The honest assessment is that $p_u(r)$ is the weakest element of the current formulation. It is a theoretical quantity with no direct observable. The four proxy tiers defined in Section 8 are operational stand-ins with varying degrees of physical motivation. The Tier I proxy is almost certainly too close to the observable to be informative. The Tier II proxy is the most defensible first candidate. The real test of the framework is whether Tier II or III proxies, applied uniformly, produce stable parameters across the SPARC sample. If they do not, the model has a proxy problem it cannot escape without additional theoretical input specifying what physical process sources p_u.

§18 Conclusion

EVM-V defines a reproducible observational programme for testing whether galactic rotation-curve residuals can be mapped onto a projection-access model derived from the IKK framework. Its primary deliverables are:

1. A precise definition of the empirical residual $a_\Pi(r) = a_{\text{obs}}(r) - a_{\text{bar}}(r)$ and a reproducible pipeline for computing it from SPARC data.
2. A minimal four-to-six-parameter velocity model with an explicit hidden-momentum proxy hierarchy and a regime gate, descended from the *Looking Up* construction.
3. A four-model comparison framework (baryons only, MOND/RAR, NFW halo, IKK residual) with AIC/BIC model-selection metrics.
4. Population-level stability tests for Λ_Π, p_0, r_c with explicit stability thresholds and correlation tests against galaxy observables.
5. Six named failure conditions that must be evaluated before any positive claim is made.
6. A minimal reproducible workflow schema specifying per-galaxy and population-level output files.

The failure conditions are not secondary. They are the primary scientific contribution. A theory that specifies how it can be shown wrong is a scientific theory. A theory that does not is a narrative. EVM-V insists on the former.

The expected next step is EVM-VI, which would apply this pipeline to the full SPARC sample and report results. Whether that paper finds stability, structured variation, or falsification is genuinely unknown. That is the point.

§A Symbol Table

Table 3: Principal symbols and their definitions

Symbol	**Meaning**	**Section**
I_a	Source identity at scale S_a	§3.1
$\mathfrak{I}_{\text{true}}$	Primitive identity (fraktur)	§3.1
$\mathcal{P}_{a\to b}$	Projection operator $S_a \to S_b$	§3.1
$\mathcal{O}_b^{(a)}$	Observable at scale S_b of source a	§3.1
$\mathfrak{R}_{a\|b}$	Projection residual $\mathfrak{I}_{\text{true}} - \widehat{I}_{a\|b}$	§3.3
r	Galactic radius	§5.1
$v_{\text{obs}}(r)$	Observed circular velocity	§5.1

(continued…)

Symbol	**Meaning**	**Section**
$\sigma_v(r)$	Velocity measurement uncertainty	§5.1
$v_{\text{bar}}(r)$	Baryonic model velocity	§5.2
$v_{\text{gas}}, v_{\text{disk}}, v_{\text{bulge}}$	Gas, disk, bulge velocity components	§5.2
$\Upsilon_{\text{disk}}, \Upsilon_{\text{bulge}}$	Stellar mass-to-light ratios	§5.2
$a_{\text{obs}}(r)$	Observed centripetal acceleration	§5.1
$a_{\text{bar}}(r)$	Baryonic Newtonian acceleration	§5.2
$a_{\Pi}(r,u)$	Projection-residual acceleration	§3.3
a_0	MOND characteristic acceleration scale	§6
$a_{\text{MOND}}(r)$	MOND acceleration	§6
$u = \ln(\ell/\ell_0)$	Logarithmic scale coordinate	§3.2
ℓ_0	Reference scale	§3.2
p_w	Legacy hidden-axis momentum (deprecated)	§3.2
$p_u(r)$	Modernised scale-access momentum proxy	§3.2
$\eta_{\Pi}(r,u)$	Projection-access efficiency	§7.3
$a_s(r)$	Scale-residual acceleration kernel	§7.3
Λ_{Π}	Projection-residual amplitude	§7.1
p_0	Hidden-momentum normalisation scale	§7.1
r_c	Core suppression radius	§7.1
r_0	Gate activation radius	§7.1
k	Gate steepness	§7.1
α_u	Proxy calibration coefficient	§8
$G_{\Pi}(r)$	Sigmoid activation gate	§7.1
$\Sigma_{\text{bar}}(r)$	Baryonic surface density	§8
$\rho_{\text{bar}}(r)$	Baryonic volume density	§8
$M_\star$	Stellar mass	§8
f_{gas}	Gas fraction	§8
R_d	Disk scale length	§8
χ^2, χ^2_ν	Chi-squared, reduced chi-squared	§10.3
AIC, BIC	Akaike, Bayesian information criteria	§10.5
S_θ	Population stability statistic	§11.1
σ_{RAR}	RAR scatter	§11.3